Task-8:- Requirement Analyss

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for setting up an **AWS RDS database** integrated with **Grafana Cloud dashboards** for infrastructure monitoring and data visualization.

This document is intended for:

- DevOps Engineers
- Cloud Engineers
- Database Administrators
- Project Mentors and Stakeholders

1.2 Scope

Included in Scope

- AWS RDS instance creation and configuration
- Database setup and dataset import
- Grafana Cloud account configuration
- Infrastructure monitoring dashboard creation
- Database analytics dashboard creation
- Performance Insights monitoring
- Query load simulation automation

Excluded from Scope

- Production-grade application deployment
- Advanced database optimization
- Multi-region disaster recovery setup

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
AWS	Amazon Web Services

RDS	Relational Database Service
DB	Database
IOPS	Input/Output Operations Per Second
SQL	Structured Query Language
VPC	Virtual Private Cloud
CloudWatch	AWS monitoring service
Grafana	Visualization and monitoring platform
Performance Insights	AWS database performance monitoring feature

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1.4 References

- AWS RDS Documentation
- AWS CloudWatch Documentation
- Grafana Cloud Documentation
- Dataset Repository:
  datasets/world.db at main · gramener/datasets

1.5 Overview

This document describes:

- Project objectives and business goals
- Functional and non-functional requirements
- Use cases and system interactions
- Risk analysis and assumptions
- Supporting glossary and appendices

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2. Project Overview

2.1 Project Objectives

- Deploy a managed relational database using AWS RDS.
- Import structured world dataset into RDS.
- Integrate AWS CloudWatch with Grafana Cloud.
- Create monitoring and analytics dashboards.
- Visualize database performance and dataset insights.

2.2 Business Goals

- Enable centralized database monitoring.
- Improve infrastructure visibility.
- Provide real-time analytics dashboards.
- Demonstrate cloud-native observability practices.
- Support performance analysis using database metrics.

2.3 Background

Modern cloud environments require monitoring solutions that combine infrastructure metrics with business-level analytics.

AWS RDS provides managed database services, while Grafana Cloud enables advanced visualization capabilities. This project integrates both platforms to simulate real-world database monitoring and analytics workflows.

2.4 Constraints

- AWS Free Tier resource limitations
- Grafana Cloud free-tier dashboard limits
- Network access configuration dependency
- Limited project implementation timeline

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3. Functional Requirements

3.1 Use Case Diagram

Actors involved:

- DevOps Engineer
- AWS RDS
- Grafana Cloud
- CloudWatch Service
- Automation Script

3.2 Use Cases

3.2.1 Use Case 1: Create AWS RDS Instance

Description:

Provision and configure an AWS RDS database instance.

Actors:

DevOps Engineer, AWS RDS

Preconditions:

- AWS account available
- Required permissions granted

Postconditions:

- RDS instance successfully running

Flow of Events:

1. Login to AWS Console
2. Create RDS instance
3. Configure networking and storage
4. Enable Performance Insights
5. Launch database instance

Alternative Flows:

- Incorrect security group blocks access
- Instance creation failure due to limits

3.2.2 Use Case 2: Import Dataset into RDS

Description:
Import world dataset into database tables.

Actors:
DevOps Engineer, Database

Preconditions:

- RDS instance active
- Database created

Postconditions:

- Tables populated with dataset

Flow of Events:

1. Connect to RDS database
2. Upload dataset
3. Execute SQL import commands
4. Validate data using queries

Alternative Flows:

- Schema mismatch
- Import failure

3.2.3 Use Case 3: Configure Grafana Cloud

Description:

Set up Grafana Cloud integrations.

Actors:

DevOps Engineer, Grafana Cloud

Preconditions:

- Grafana Cloud account created

Postconditions:

- Data sources successfully connected

Flow of Events:

1. Login to Grafana Cloud
2. Add CloudWatch datasource
3. Add RDS database datasource
4. Test connectivity

3.2.4 Use Case 4: Create Infrastructure Monitoring Dashboard

Description:

Monitor RDS infrastructure metrics.

Actors:

Grafana Cloud, CloudWatch

Preconditions:

- CloudWatch integration configured

Postconditions:

- Dashboard displays infrastructure metrics

Flow of Events:

1. Create dashboard
2. Add CPU utilization panel
3. Add memory usage panel
4. Add IOPS and connections metrics

3.2.5 Use Case 5: Create Database Analytics Dashboard

Description:

Visualize dataset insights using SQL queries.

Actors:

Grafana Cloud, Database

Preconditions:

- Dataset available in RDS

Postconditions:

- Analytics visualizations generated

Flow of Events:

1. Execute SQL queries
2. Fetch aggregated data
3. Create visualization panels

3.2.6 Use Case 6: Database Load Simulation (Bonus)

Description:

Generate database workload automatically.

Actors:

Automation Script, Database

Preconditions:

- Database accessible

Postconditions:

- Query load visible in Performance Insights

Flow of Events:

1. Execute random queries
2. Schedule cron job
3. Monitor DB load metrics

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4. Non-Functional Requirements

4.1 Performance Requirements

- Dashboard refresh interval ≤ 30 seconds
- SQL queries executed within acceptable latency
- System supports continuous monitoring

4.2 Security Requirements

- Restricted database access via security groups
- Secure credential storage
- Encrypted database connection

4.3 Usability Requirements

- Easy-to-understand dashboards
- Clear visualization panels
- Minimal manual configuration

4.4 Reliability Requirements

- High availability via AWS managed RDS
- Continuous metric collection
- Automatic monitoring recovery

4.5 Compliance Requirements

- Follow AWS security best practices
- Secure handling of database credentials

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5. Assumptions and Dependencies

5.1 Assumptions

- AWS services are available.
- Internet connectivity exists.
- Dataset format compatible with DB engine.

5.2 Dependencies

- AWS CloudWatch service
- Grafana Cloud platform
- Network connectivity between services

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6. Risk Analysis

6.1 Risk Identification

- Incorrect security configuration
- Database connectivity issues
- Free-tier resource exhaustion
- Integration failures

6.2 Risk Mitigation

- Validate security group rules
- Test connectivity regularly
- Monitor resource utilization
- Maintain backup configurations

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7. Glossary

Term	Meaning
Dashboard	Visual representation of metrics
Datasource	System providing monitoring data
Metric	Measurable system parameter
Query Load	Database processing workload

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8. Appendices

- AWS Architecture screenshots
- Grafana Dashboard screenshots

- SQL Queries used for analytics
- Automation script details

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9. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____